A TAUT MAP FROM A PRE-ADIC RING WITH FINITELY GENERATED IDEAL OF DEFINITION TO A COMPLETE RING IS ADIC

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays no diagram; its content is prose and inline formulas, reproduced verbatim.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The completion with respect to a finitely generated ideal, and its quotients. Algebra, Lemma algebra-lemma-hathat-finitely-generated, invoked at proof lines 1569-1572. Identifies the quotients of the adic completion of the target with the quotients of the target.
- A continuous taut map with dense image is open and surjective. Same chapter, Lemma lemma-dense-image-surjective, invoked at proof lines 1575-1577. Gives that the topology of the target is the adic one.
- Tautness of the given map, and completeness of the target. Same chapter, the definitions in force in the statement. Supply the identifications used in the comparison.

Source reference. Original lemma 05GG: sources/stacks-chow/tags.txt, source lines 7060–7060. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0APT: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1455–1455. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $\varphi : A \to B$ be a continuous map of linearly topologized rings. If φ is taut, A is pre-adic and has a finitely generated ideal of definition, and B is complete, then B is adic and has a finitely generated ideal of definition and the ring map φ is adic.*

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Choose a finitely generated ideal of definition I of A. Let J_n be the closure of $\varphi(I^n)B$ in B. Since B is complete we have $B = \lim B/J_n$. Let $B' = \lim B/I^nB$ be the I-adic completion of B. By Algebra, Lemma 05GG, the I-adic topology on B' is complete and $B'/I^nB' = B/I^nB$. Thus the ring map $B' \to B$ is continuous and has dense image as $B' \to B/I^nB \to B/J_n$ is surjective for all n. Finally, the map $B' \to B$ is taut because $(I^nB')B = I^nB$ and $A \to B$ is taut. By Lemma 0APT we see that $B' \to B$ is open and surjective. Thus the topology on B is the I-adic topology and the proof is complete. $\square$